COUNTRY PROFILES

Uranium in Uzbekistan

UPDATED WEDNESDAY, 11 FEBRUARY 2026

- Uzbekistan has considerable mineral deposits, including uranium.
- It is the world's fifth-ranking uranium supplier.
- Construction of two large and two small Russian reactors is planned.

Uzbekistan was a significant source of Russian uranium supply until independence in 1991. Uranium production until then took place in Uzbekistan, Kyrgyzstan and Tajikistan, with little regard for national borders and much of the treatment being in Tajikistan.

Privacy - Terms

Today, most uranium is mined in the middle of the country, with Navoi as the centre, linked to mines by railway.

Uzbekistan is planning to build a nuclear power plant utilizing large and small Russian-designed reactors.

Electricity sector

Total generation (in 2023): 74.3 TWh

Generation mix: natural gas 56.4 TWh (76%); coal 8.3 TWh (11%); hydro 6.9 TWh (9%); oil 2.1 TWh (3%).

Import/export balance: 2.8 TWh net import (4.4 TWh imports; 1.7 TWh exports)

Total consumption: 54.6 TWh

Per capita consumption: c. 1500 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Electricity is imported from Tajikistan and Kyrgyzstan. The country is committed to reducing consumption of natural gas for power generation to free it up for higher-value purposes, including the petrochemicals industry. Accordingly, it plans to build nuclear power capacity to generate about 15% of power.

Plans for power reactors

In December 2017 Uzbekistan and Russia signed an intergovernmental agreement on cooperation in the use of nuclear energy for peaceful purposes. In March 2018 Uzbekistan shortlisted 10 sites across the country for construction of a nuclear power plant, and by October the focus was on sites on the eastern and western shores of Lake Tudakul, a reservoir 25km northeast of Bukhara, on the border of the provinces of Navoi and Bukhara. In September 2018 a further intergovernmental agreement was signed for construction by Rosatom of two VVER-1200 reactors to be commissioned about 2028. The reference units are Novovoronezh II (V-392M).

Rosatom in May 2019 said that a site near Aydarkul Lake in the Jizzakh region had been chosen, near the north-central border with Kazakhstan and north of Samarkand, and that construction was

expected to start in 2020 or 2021. In July 2019 it was reported that the proposed site, would be near Lake Tuzkan, in the Farishsky district. UzAtom's site evaluation is being assisted by the International Atomic Energy Agency (IAEA) as part of its 'Milestones' approach to establishing nuclear power.

In June 2019 French-based engineering and project management group Assystem signed an agreement with the Ministry of Energy to help double the country's installed generating capacity from 13 to 27 GWe and double its output by 2030, using nuclear, gas and wind.

In May 2020 the country's Ministry of Energy published a 'Concept Note' outlining its strategy for electricity generation through 2030. The document forecasted 15% of the country's electricity to come from nuclear by 2030, as well as 8% from solar and 7% from wind.

In May 2021 the IAEA conducted an integrated nuclear infrastructure review (INIR) to assess nuclear infrastructure development in the country.

In January 2023 the IAEA completed a five-day Site and External Events Design mission reviewing Uzbekistan's safety processes for evaluating the site of its first nuclear power plant. The review found that Uzbekistan had "carried out an objective and safety-oriented site characterization process that emphasized the safety of workers, public and environment, in accordance with the IAEA Safety Standards."

In May 2024, Uzbekistan and Russia signed a contract to build an SMR plant in the Jizzakh region. The plan was initially for six 55 MWe RITM-200N reactors. However, in June 2025 an agreement was signed for a plant featuring two or four VVER-1000 reactors, and the contract for SMRs was adjusted to two units. In September 2025 a further agreement was signed confirming the plan for two VVER-1000 units and two RITM-200N units.

Earlier in April 2025 preparatory work began on auxiliary buildings at the site, and in May Uzatom approved the site layout for the plant.

In September 2025 Uzatom signed a deal with Rosatom to supply nuclear fuel for the country's planned nuclear power plants, and in October excavation works began for the RITM-200N reactors. First concrete for the the first RITM-200N unit is expected in 2026, with criticality expected in late 2029.

The two VVER-1000 units are expected to be commissioned in 2033 and 2035.

In November 2025, during President Mirziyoyev's visit to the USA, Uzbekistan expressed interest in studying American SMR technologies from Holtec, GE Hitachi and NuScale alongside the Russian programme. In January 2026 Uzatom signed a scientific and technical cooperation agreement with Russia's Kurchatov Institute, covering small modular and mobile reactor development, nuclear medicine and specialist training.

Planned nuclear power reactors in Uzbekistan

Region	Type	MWe gross
Jizzakh	RITM-200N	2x55
Jizzakh	VVER-1000	2x1200

Uranium resources

According to the 2024 *Red Book*, Uzbekistan has 40,200 tU in reasonably assured recoverable resources and 49,220 tU inferred recoverable resources to $130/kgU.

All significant sandstone roll-front-type uranium resources are located in the Central Kyzylkum area, a 125 km-wide belt extending over a distance of about 400 km from Uchkuduk in the northwest, to Nurabad in the southeast.

In April 2015 NMMC announced plans approved by the government to implement 27 projects to modernize its production facilities by 2019, at a total cost of $985 million. Among the projects were the construction of a mining and distribution complex in Samarkand region, the development of the main raw material base – the Muruntau open pit mine, and the modernization and technical re-equipment of other production facilities. NMMC's four metallurgical plants in Navoi, Zarafshan, Uchkuduk and Zarmitan were mentioned.

In September 2019 Orano signed an agreement with Uzbekistan for cooperation in the development of uranium mining and exploration. A joint venture, Nurlikum Mining, was set up between Orano (51%) and Goscomgeology (the State Committee for Geology and Mineral Resources of the Republic of Uzbekistan, 49%) to focus on the Djengeldi areas of Kyzylkum. The objective of

Nurlikum Mining is to develop the project to mine the Djelgendi uranium deposit. In November 2022 a strategic cooperation agreement was signed by Orano Mining, Goscomgeology and Uzbek state-owned enterprise Navoiuran to implement several joint initiatives in Uzbekistan, including geological exploration projects and development of new uranium mines, whilst also working on a roadmap for the development of the Djengeldi project. In November 2023, Orano reported that it had extracted its first 350 kgU from the deposit.

In March 2025 Japanese company Itochu Corporation joined the Nurlikum Mining joint venture. Orano and Navoiyuran each now hold a 45% share, with 10% held by Itochu. In October 2025 the partners completed the restructuring of the joint venture, with Navoiyuran replacing Goscomgeology as the Uzbek partner and serving as project operator. The first phase of the South Djengeldi project involves investment of $214 million over three years, with average annual output expected to be about 500 tU.

Earlier in July 2022 Uzbekistan's President signed a resolution to increase the country's uranium production from 3500 tU to 7100 tU by 2030. In December 2025 President Mirziyoyev instructed officials to finalize the feasibility study for a newly identified uranium deposit at Arnasay in the Forish district, Jizzakh region.

Navoi Mining & Metallurgical Company (NMMC) is part of the Uzbekistani state holding company Kyzylkumredmetzoloto, and undertakes all uranium mining in the country, as well as gold mining and other activities. From the beginning of 2022, NMMC was split into three independent entities: Navoi Mining and Metallurgical Plant (production of precious metals); Navoiuran (mining and processing of natural uranium and rare earth metals); and Navoi Mining and Metallurgical Plant's Foundation.

History and export of uranium

Uranium mining commenced in 1953 at a deposit near Uchkuduk. NMMC was set up in 1958 and focused on uranium and gold in the Central Kyzylkum/Qizilqum desert region of Navoi province, particularly the Uchkuduk deposit (mined underground and open pit from 1964), which led to the discovery of subsequent similar deposits. In 1966, underground mining began at the Sabyrsay deposit in Samarkand and in 1977 at the Sugraly deposit near Zarafshan. In 1978, ISL was initiated at the Ketmenchi deposit. In 1975 the Sabyrsay deposit was being mined by ISL. Underground

mining continued to 1990 and open pit to 1994, but mines are all now in situ leach (ISL). Uranium-bearing solutions are sent by rail to the central hydrometallurgical HMP-1 plant in Navoi for final recovery of U_3O_8 product. This plant was commissioned in 1964 and developed through to 1983. Bacterial leaching was introduced in 2011.

Five "company towns" were constructed to support uranium production activities: Uchkuduk, Zarafshan, Zafarabad, Nurabad, and Navoi, with a combined population of some 500,000. They remain centres of five mining districts.

Until 1992, all uranium mined and milled in Uzbekistan was shipped to Russia. Since 1992, practically all Uzbekistani uranium production has been exported to the USA and other countries through Nukem Inc.

In 2008 South Korea's Kepco signed agreements to purchase 2600 tU over six years to 2015, for about US$ 400 million. In May 2014 China's CGN agreed to buy $800 million of uranium through to 2021, and China customs was reported as saying that Uzbekistan was second only to Kazakhstan as a uranium supplier to the country.

Uzbekistan has historically held uranium production information as confidential. The table below provides estimates of annual uranium production between 2015 and 2024.

Estimated Uzbek uranium production (tonnes U)

Division	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Total	2385	3325	3400	3450	3500	3500	3516	3561	4000	4000

NMMC has several divisions, the Northern and Central ones plus part of #5 are in the Kyzylkum Desert region of Navoi province.

NMMC divisions

Northern mining directorate

Centred on **Uchkuduk**, the Northern mining district 300 km north of Navoi was established to mine uranium at Uchkuduk, from 1961, by underground and open pit mines, with ore treated at the central plant in Navoi. Since 1965 ISL uranium mining has been used at

Uchkuduk and since 1995, at Kendyk-Tyube. There is also a 450,000 t/yr sulfuric acid plant at Uchkuduk (possibly in conjunction with a copper smelter).

Central mining directorate

In the **Zarafshan** or eastern mining district, about 200 km north of Navoi, Sugraly was mined underground from 1977 and then ISL to 1994, when it was closed. NMMC had a joint venture with Areva (now Orano) to redevelop the **Sugraly** deposit with reported 38,000 tU resources, but this has lapsed. Sugraly is a thick deposit with complex mining and geological conditions and high carbonate content. An integrated processing facility at Sugraly was put into operation in 2014.

Mining directorate #5

The mining district #5 mostly in Bukhara province, west of Navoi and headquartered in **Zafarabad**, close to Navoi was set up in 1971 by another entity in Bukhara province and became part of NMMC in 1993. It mines the Bukinay group of uranium deposits by ISL methods. Mines include North & South Bukinay (from 1970), Beshkak (from 1978), Istiklol, Kukhnur, Lyavlyakan (from 1998), Tokhumbet (from 2004) and South Sugraly.

Southern mining directorate

The Southern mining district in Samarkand province, southeast of Navoi, and headquartered at **Nurabad**, was founded in 1964 to mine the Sabirsay uranium deposit by underground methods, which continued to 1983. ISL then took over, and continues to be the main mining method. The operation was transferred from Tajikistan to NMMC about 1994*. Other mines are Ketmenchi (ISL since 1978), Jaarkuduk, Yogdu, Shark and Ulus.

* In Soviet times, to 1992, the Leninabad Mining and Chemical Combine located at Chkalovsk, a few kilometres southeast of Khujand (Khodjend/Khodzhent), Sughd province, northeren Tajikistan, on the Syr Darya River in the western part of the Ferghana Valley, is to the east of Uzbek's Samarkand province. The Combine incorporated seven mines and several plants, notably Combine No. 6 (Uranium Plant V), and it processed up to 1,000,000 tonnes of uranium ore per year to produce yellowcake for the Soviet nuclear power and defence industries. A lot of this was at the central plant. Reportedly,

Chkalovsk once had the capability to convert uranium concentrate into uranium hexafluoride. It was established in 1945 as a large-scale hydrometallurgical uranium enterprise, based on the uranium deposits of Tajikistan, Kyrgyzstan and Uzbekistan. Uranium mining ceased in Tajikistan in 1992, and the Combine became the industrial association Vostochnyy Eastern Combine for Rare Metals (IA Vostokredmet) which is a significant industrial complex specialising today in underground and heap leaching of metals. There are considerable amounts of uranium tailings and disturbed areas from the combine's operation – over 100 square kilometres overall in Tajikistan, according to a Russian estimate in a 2002 IAEA report. See Appendix.

Mining directorate #2

In MA#2 at Krasnogorsk, Parkent region, Tashkent province in the east of the country, the Chauly uranium deposit was previously mined, but the focus there is on phosphorite now. It became part of NMMC in 1995.

New mines: Northern, Central and #5

NMMC in 2008 started mining the major new **Northern Kanimekh** deposit, northwest of Navoi, costing $34 million. Northern Kanimekh ore occurs 260 to 600 metres deep with 77% of uranium reserves present at 400-500-metre depth. This requires a special approach to building wells and uranium mining process. The pilot plants at Northern Kanimekh and Alendy (Directorate #5) were commissioned in 2008-09 and the two commercial mines were completed at the end of 2013.

NMMC also built a pilot plant for ISL at Yarkuduk and Tokhumbet deposits. It has started operation of the $21 million Aulbek ISL mine near Lyavlyakan in central Kyzylkum (Directorate #5), which ramped up to 2013, and also Meylisay and Tutlinskaya ploshchad, costing about $30 million.

Over 2008-12 NMMC invested $230 million in upgrades to expand the existing mining and processing capacities, renew the fleet of process equipment, and establish up to seven new mines. "As part of an increase in uranium production up to 2012, the expansion and reconstruction of sulfuric acid production, at a cost of about $12 million, will be carried out. Implementation of the program will make it possible to increase uranium production in 2012 by 50%". Early in 2009 the Uzbek president said that the world economic crisis would

slow all this development, though full commissioning of the Aulbek, Alendy and North Kanimekh mines in Central Kyzylkum at the end of 2013 would enable a 40% increase of NMMC production.

In August 2013 NMMC suspended construction of Meylisay and Northern Maizak mines in central Kyzylkum due to high carbonate content in the ore rendering ISL inefficient. However Navoi said that in 2015 the technology of uranium mining would be optimized at Meylisay deposit. Aulbek and North Kanimekh also have high carbonate levels.

According to IAEA's 2024 Red Book, in 2022, 18 uranium deposits were in operation by ISL method and 4 more deposits are planned for development during 2022-2026.

International ventures, black shales

Due to the complexity of required processing and high projects costs, development of black shales has been delayed indefinitely.

China: In August 2009 Goscomgeo and China Guangdong Nuclear Uranium Corp. (CGN-URC) set up a 50-50 uranium exploration joint venture, Uz-China Uran, to focus on the black shale deposits in the Boztau-skaya area in the central Kyzylkum desert of the Navoi region. Some 5500 tU resources are reported. Over 2011-13 CGN-URC was to develop technology for the separate production of uranium and vanadium from these black shale deposits with a view to commencing production from them, but no activities have been reported since then.

Russia: In January 2006 Techsnabexport (ARMZ subsidiary) signed a memorandum of understanding with NMMC and Goskomgeo to set up a uranium mining joint venture based on the Aktau deposit. Initially, it was planned that the joint venture would start operations late 2006, but after four years' negotiation no agreement could be reached and Russia withdrew in mid-2010. The ore is complex and this has apparently deterred establishment of the project. Goskomgeo invited ARMZ to consider its black shales, but ARMZ declined on the basis that no treatment process was known for them.

In 2007 Russia offered to enrich Uzbekistan uranium in the International Uranium Enrichment Center in Angarsk.

Japan: In September 2006 a Japan-Uzbek intergovernmental agreement was aimed at financing Uzbek uranium development and in October 2007 Itochu Corporation agreed with NMMC to

develop technology to mine and mill the black shales, particularly the Rudnoye deposit, and to take about 300 tU/yr from 2007. A 50-50 joint venture was envisaged, but no more was heard until February 2011 when Itochu signed a 10-year "large-scale" uranium purchase agreement with NMMC.

In mid-2008 Mitsui & Co. signed a basic agreement with the Uzbek government's Goscomgeo to establish a joint venture for geological investigations regarding the development of black-shale uranium resources at the Zapadno-Kokpatasskaya mine, 300 km NW of Navoi.

In mid-2009, and further to an April 2007 MOU, Goscomgeo and the Japan Oil, Gas and Metals National Corporation (JOGMEC) signed an agreement for uranium and rare earths exploration in the Navoi region, focused on ISL-type sandstone deposits and black shales, with a view to a Japanese company taking a 50% interest in any resources identified and developing them. In February 2011 a further broad agreement was signed between the two. In August 2013 JOGMEC was granted a licence to explore for uranium in two sandstone deposits, Juzkuduk and Tamdiykuduk-Tulyantash, for five years with rights to mine, following a July agreement with NMMC.

Organization

The Uzbekistani State Committee for Safety in Industry and Mining (Gosgortekhnadzor) supervises ministries engaged in mining.

The Nuclear Regulations Inspectorate under Gosgortekhnadzor has responsibility for the control and supervision of the research reactors and all nuclear and radioactive materials (including spent fuel) in Uzbekistan.

In June 2017 the government set up a new centralized directorate for geological exploration for uranium, precious, and non-ferrous metals. The scientific and production centre "Geology of Precious Metals and Uranium" of Navoiy Mining and Metallurgy Combinat (NMMC) and the scientific and production centre "Geology of non-ferrous metals" of the Almalyk Mining and Metallurgical Complex are to be transferred to this.

The Nuclear Energy Development Agency (Uzatom) was set up in July 2018 and collaborates with Rosatom and the Uzbek Academy of Sciences and is responsible for the development of relevant infrastructure and training for the country's nuclear power

programme. In May 2024 Uzatom was restructured as the Atomic Energy Agency under the Cabinet of Ministers.

Russia's regulator, Rostechnadzor, and the International Atomic Energy Agency (IAEA) will help set up the new independent Uzbek regulator, and on IAEA advice Russia's standards and regulations will be the basis of Uzbekistan's.

At the beginning of 2022, NMMC became three independent entities: Navoi Mining and Metallurgical Plant; Navoiuran; and Navoi Mining and Metallurgical Plant's Foundation.

Research & development

There have been two research reactors, a 10 MW tank type – WWR-SM – operating since 1959 at the Institute of Nuclear Physics, Uzbek Academy of Sciences near Tashkent, and a small 20 kW one operated by JSC Foton in Tashkent. This was converted in 2008 to run on low-enriched uranium from TVEL, and all HEU fuel was returned to Russia. Decommissioning the Foton reactor was undertaken over 2015-19. The larger WWR-SM shut down in July 2016, with decommissioning intended to begin soon after. However, in February 2017 it was decided to refurbish it, and restart it in July 2017. In March 2018 an IAEA peer-review mission visited at the request of the government and made recommendations related to its long-term operation. In November 2024 the IAEA conducted a Safety Aspects of Long-term Operation (SALTO-RR) peer review of the reactor, recommending formal periodic safety reviews and an updated ageing management programme. A follow-up mission is planned for 2027.

The state enterprise Scientific Production Centre Urangeologiya undertakes uranium exploration in new areas.

Legacy sites

See section in information papers on Tajikistan and Kyrgyzstan.

In June 2015 the European Bank for Reconstruction and Development (EBRD) set up a fund to deal with radioactive contaminated material resulting from Soviet-era uranium mining and processing in the Kyrgyz Republic, Tajikistan and Uzbekistan. It drew attention to legacy sites along the tributaries to the Syr Darya River, running through the Fergana Valley, which is shared by the Kyrgyz Republic, Tajikistan and Uzbekistan.

Non-proliferation

Uzbekistan is a party to the NPT since 1994 and in 1998 ratified an Additional Protocol agreement with the IAEA. It has also ratified the Central Asia Nuclear Weapon Free Zone treaty, with Kyrgyzstan and Tajikistan.

Notes & references

General references

OECD NEA & IAEA, 2024, *Uranium 2024: Resources, Production and Demand* ('Red Book')

NMMC website

Burykin, A.A., Iskra, A.A., Karamushka, V.P., Radiation Legacy of the USSR Enterprises for Mining, Milling and Processing of Uranium Ores: Conservation, Decommissioning and Environmental Rehabilitation, p244-256 of Radiation legacy of the 20th century: Environmental restoration, Proceedings of an International Conference (RADLEG 2000) held in Moscow, Russian Federation, 30 October-2 November 2000 and organized by the Ministry of the Russian Federation for Atomic Energy in co-operation with the International Atomic Energy Agency, the European Commission and the Russian Academy of Sciences, April 2002, IAEA-TECDOC-1280

Egorov, N.N., Novikov, V.M., Parker F.L., Popov V.K., *The Radiation Legacy of the Soviet Nuclear Complex: An Analytical Overview*, International Institute of Applied Systems Analysis (first published 2000)

Related information

World Uranium Mining Production